EDP Supplement

University of Illinois at Chicago

2023 ChemE Car Team

Captain: Ramiro Maldonado

Advisor: Meenesh Singh

Table of Contents

Safety Certificates……3-7

Water……8

Sulphuric Acid……...14

Lead Sulphate…….....22

Lead Plate……...25

Hydrogen Peroxide…….…...33

Acetic Acid……42

Indium(III) iodide……..50

Potassium Iodide……57

Sodium acetate, anhydrous……63

Sodium thiosulfate pentahydrate, ACS……..69

Starch, soluble……79

Arduino Uno……..98

FA-130RA Motor……..99

Acrylite Acrylic……...100

PLA……..10

AIChE ACADEMY

This Certificate of Completion certifies that

Mario Ocampo

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with Professional Development Hours [PDHs] which is equivalent to Continuing Education Units [CEUs].

Date Awarded:
February 2, 2022

June C. Wispelwey, Executive Director & CEO, AIChE®

AIChE Academy
Connect | Learn | Solve

This certifies that

Ramiro Roman

Has successfully completed the eLearning course:

AIChE's ChemE Car Safety Course

Date Awarded:

September 22, 2020

Dr. Skip Rochefort
Chem-E-Car Competition Chair

AIChE ACADEMY

This certifies that

Ramiro Maldonado

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs]
which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
July 25, 2022

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This Certificate of Completion certifies that

Vihar Trada

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with Professional Development Hours [PDHs]
which is equivalent to Continuing Education Units [CEUs].

Date Awarded:
October 26, 2021

June C. Wispelwey, Executive Director & CEO, AIChE®

AIChE ACADEMY

This certifies that

Emily Seriruk

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs]
which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
November 29, 2022

Darlene Schuster, Executive Director & CEO, AIChE®

AIChE ACADEMY

This certifies that

Andrew Russ

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
January 26, 2023

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Karim Al Al Zahabi

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
July 25, 2022

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Jocelyn Alinn Ortega

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
January 31, 2023

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Ogechikanma Ihenacho

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
February 3, 2023

Darlene Schuster, Executive Director & CEO, AIChE

AIChE ACADEMY

This certifies that

Varun Maheshwari

Has successfully completed:

ELA199 ChemE Car Safety Course

And is therefore credited with 0.00 Professional Development Hours [PDHs] which is equivalent to 0.00 Continuing Education Units [CEUs].

Date Awarded:
February 1, 2023

Darlene Schuster, Executive Director & CEO, AIChE

Powered by TCPDF (www.tcpdf.org)

SAFETY DATA SHEET

Creation Date 26-Jan-2010 **Revision Date** 24-Dec-2021 **RevisionNumber**9

1. Identification

Product Name Water

Cat No. : OPTIMAKIT; W2-4; W2-20; W2SS-200; NC1188975; W11-1; W11-4;XXBA192

CAS No 7732-18-5
Synonyms (Deionized/Ultra-Filtered/RNA/DEPC-Treated/Nuclease-Free/Molecular Biology/DNA/Biotech/HPLC/Plasma/Environmental/Optima LC/MS)

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company
Fisher Scientific Company
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number CHEMTRECÒ, Inside the USA: 800-424-9300 CHEMTRECÒ, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification
Classification under 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1

Label Elements
None required

Hazards not otherwise classified (HNOC)
None identified

3. Composition/Information on Ingredients

Component	CAS No	Weight %
Water	7732-18-5	100

4. First-aid measures

Eye Contact No hazards which require special first aid measures. **Skin Contact** No hazards which require special first aid measures. **Inhalation** No hazards which require special first aid measures. **Ingestion** No hazards which require special first aid measures. No information available.

Most important symptoms and effects
Notes to Physician Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media Substance is nonflammable; use agent most appropriate to extinguish surroundingfire. **Unsuitable Extinguishing Media** No information available

Flash Point Not applicable
Method - No information available

Autoignition Temperature No information available
Explosion Limits
Upper No data available
Lower No data available
Sensitivity to Mechanical Impact No information available
Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical
Non-combustible. None reasonably foreseeable.

Hazardous Combustion Products
None under normal use conditions.
Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear.

NFPA

Health	Flammability	Instability	Physical hazards
0	0	0	N/A

6. Accidental release measures

Personal Precautions If spilled, take caution, as material can cause surfaces to become very slippery. **Environmental Precautions** No special environmental precautions required.

Methods for Containment and Clean Up Soak up with inert absorbent material.

7. Handling and storage

Handling No special handling advices are necessary.

Storage. No special storage conditions required. Incompatible Materials. None known.

8. Exposure controls / personal protection

Exposure Guidelines This product does not contain any hazardous materials with occupational exposurelimitsestablished by the region specific regulatory bodies.

Engineering Measures None under normal use conditions.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166.

Skin and body protection No special protective equipment required.

Respiratory Protection No protective equipment is needed under normal use conditions. Handle in accordance with good industrial hygiene and safety practice.

Hygiene Measures

9. Physical and chemical properties

Physical State Liquid
Appearance Clear, Colorless
Odor Odorless
Odor Threshold No information available **pH** 7
Melting Point/Range 0 °C / 32 °F **Boiling Point/Range** 100 °C / 212 °F **Flash Point** Not applicable
Evaporation Rate No information available **Flammability (solid,gas)** Not applicable
Flammability or explosive limits
Upper No data available
Lower No data available
Vapor Pressure 17.5 mmHg @ 20 °C **Vapor Density** No information available **Specific Gravity** 1.000
Solubility miscible
Partition coefficient; n-octanol/water No data available **Autoignition Temperature** No information available
Decomposition Temperature No information available **Viscosity** No information available **Molecular Formula** H2O
Molecular Weight 18.02

10. Stability and reactivity

Reactive Hazard None known, based on information available **Stability** Stable under normal conditions.

Conditions to Avoid Incompatible products. Excess heat.

Incompatible Materials None known

Hazardous Decomposition Products None under normal use conditions

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information Product does not present an acute toxicity hazard based on known information**Component Information**

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-

No information available

Toxicologically Synergistic Products

Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** No information available

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.

Component	CAS No	IARC	NT	ACGIH	OSH	Mexi
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known
STOT - repeated exposure None known

Aspiration hazard No information available

No information available

Symptoms / effects,both acute and delayed

Endocrine Disruptor Information No information available

Other Adverse Effects No information available.

12. Ecological information

Ecotoxicity
Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

Persistence and Degradability Miscible with water Persistence is unlikely based on information available. **Bioaccumulation/**

Accumulation No information available.

Mobility Will likely be mobile in the environment due to its water solubility.

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and

national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT Not regulated
TDG Not regulated
IATA Not regulated
IMDG/IMO Not regulated

15. Regulatory information

United States of America Inventory

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	TSCA - EPA RegulatoryFlags
Water	7732-18-5	X	ACTIVE	-

Legend:
TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)
X - Listed
'-' - Not Listed

TSCA 12(b) - Notices of Export Not applicable

International Inventories
Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea(KECL).

Component	CAS No		DSL	NDSL	EINECS	PI	E	I	AIC	IECSCKEC
Water	7732-18-5		X	231-791-2						XKE-35400

KECL - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

U.S. Federal Regulations

SARA 313 Not applicable

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act) Not applicable

Clean Air Act Not applicable

Not applicable

OSHA - Occupational Safety and Health Administration

CERCLA Not applicable

California Proposition 65 This product does not contain any Proposition 65 chemicals. Not applicable

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Water	-	-	X	-	-

U.S. Department of Transportation
Reportable Quantity (RQ): N
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

This product does not contain any DHS chemicals.

U.S. Department of Homeland Security

Other International Regulations

Mexico - Grade No information available

Authorisation/Restrictions according to EU REACH

Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone DepletionPotential	Restriction of Hazardous Substances (RoHS)

Water	7732-18-5	Listed	Not applicable	Not applicable	Not applicable

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantitiesfor Major AccidentNotification	Seveso III Directive(2012/18/EC) - Qualifying Quantitiesfor Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention(Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable

16. Other information

Prepared By Regulatory Affairs

Thermo Fisher Scientific

Email: EMSDS.RA@thermofisher.com

Creation Date 26-Jan-2010
Revision Date 24-Dec-2021
Print Date 24-Dec-2021

Revision Summary This document has been updated to comply with the US OSHA HazCom2012Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information andbelief at thedate of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,transportation, disposal and release and is not to be considered a warranty or quality specification. The informationrelates only to the specific material designated and may not be valid for such material used in combinationwithanyothermaterials or in any process, unless specified in the text

End of SDS

1. Identification

Product Name Sulfuric acid

Cat No. : 11000

Synonyms Hydrogen sulfate; Vitriol brown oil; Oil of vitriol

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.
Details of the supplier of the safety data sheet

Company
Alfa Aesar
Thermo Fisher Scientific Chemicals, Inc.
30 Bond Street
Ward Hill, MA 01835-8099
Tel: 800-343-0660
Fax: 800-322-4757
Email: tech@alfa.com
www.alfa.com

Emergency Telephone Number
During normal business hours (Monday-Friday, 8am-7pm EST), call (800) 343-0660.
After normal business hours, call Carechem 24 at (866) 928-0789.

2. Hazard(s) identification

Classification
This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Corrosive to metalsCategory 1
Skin Corrosion/Irritation Category 1 A Serious Eye Damage/Eye Irritation Category 1 Specific target organ toxicity (single exposure) Category 3 Target Organs - Respiratory system.

Label Elements

Signal Word
Danger

Hazard Statements
May be corrosive to metals
Causes severe skin burns and eye damage
May cause respiratory irritation



Precautionary Statements
Prevention
Do not breathe dust/fume/gas/mist/vapors/spray
Wear protective gloves/protective clothing/eye protection/face protection
Wash face, hands and any exposed skin thoroughly after handling
Use only outdoors or in a well-ventilated area
Keep only in original container
Response
Immediately call a POISON CENTER or doctor/physician
Inhalation
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing **Skin**
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower Wash contaminated clothing before reuse
Eyes
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continuerinsing**Ingestion**
IF SWALLOWED: Rinse mouth. DO NOT induce vomiting
Storage
Store locked up
Store in a well-ventilated place. Keep container tightly closed
Disposal
Dispose of contents/container to an approved waste disposal plant
Hazards not otherwise classified (HNOC)
None identified

WARNING. Cancer - https://www.p65warnings.ca.gov/.

3. Composition/Information on Ingredients

Component	CAS-No	Weight %
Sulfuric acid	7664-93-9	90 - 98
Water	7732-18-5	2 - 10

4. First-aid measures

General Advice Show this safety data sheet to the doctor in attendance. Immediate medical attentionisrequired.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Remove andwashcontaminated clothing and gloves, including the inside, before re-use. Immediatemedical attention is required.

Inhalation Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouthmethod if victim ingested or inhaled the substance; give artificial respiration withtheaidofa pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

Ingestion Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Call aphysician or poison control center immediately.

Most important symptoms and effects

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagusshouldbe investigated: Ingestion causes severe swelling, severe damage to the delicatetissueand danger of perforation

Notes to Physician Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media CO_2, dry chemical, dry sand, alcohol-resistant foam. **Unsuitable**

Extinguishing Media DO NOT USE WATER

Flash Point Not applicable
Method - No information available

Autoignition Temperature No information available
Explosion Limits
Upper No data available
Lower No data available
Sensitivity to Mechanical Impact No information available
Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical
Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin andmucousmembranes.

Hazardous Combustion Products
Sulfur oxides. Hydrogen.
Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

NFPA

Health	Flammability	Instability	Physical hazards
3	0	2	W

6. Accidental release measures

Personal Precautions Ensure adequate ventilation. Use personal protective equipment as required. Evacuatepersonnel to safe areas. Keep people away from and upwind of spill/leak.
Environmental Precautions Should not be released into the environment.

Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

7. Handling and storage

Handling Wear personal protective equipment/face protection. Do not get in eyes, on skin, or onclothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Donot ingest. If swallowed then seek immediate medical assistance.

Storage Keep containers tightly closed in a dry, cool and well-ventilated place. Keep awayfromwater or moist air. Corrosives area.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Sulfuric acid	TWA: 0.2 mg/m^3	(Vacated) TWA: 1 mg/m^{3}TWA: 1 mg/m^3	IDLH: 15 mg/m^3 TWA: 1 mg/m^3	TWA: 0.2 mg/m^3

Legend

ACGIH *- American Conference of Governmental Industrial Hygienists*
OSHA *- Occupational Safety and Health Administration*
NIOSH IDLH: *NIOSH - National Institute for Occupational Safety and Health*

Engineering Measures Use only under a chemical fume hood. Ensure adequate ventilation, especially inconfinedareas. Ensure that eyewash stations and safety showers are close to the workstation location.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166.

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or EuropeanStandardEN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Handle in accordance with good industrial hygiene and safety practice.

Hygiene Measures

9. Physical and chemical properties

Physical State Liquid
Appearance Clear, Colorless to brown **Odor** Odorless
Odor Threshold No information available **pH** 0.3 (1N)
Melting Point/Range 10 °C / 50 °F **Boiling Point/Range** 290 - 338 °C / 554 - 640.4 °F **Flash Point** Not applicable
Evaporation Rate Slower than ether
Flammability (solid,gas) Not applicable
Flammability or explosive limits
Upper No data available
Lower No data available
Vapor Pressure < 0.001 mmHg @ 20 °C **Vapor Density** 3.38 (Air = 1.0)
Specific Gravity 1.84
Solubility Soluble in water
Partition coefficient; n-octanol/water No data available **Autoignition Temperature** No information available
Decomposition Temperature 340°C
Viscosity No information available **Molecular Formula** H2SO4

Molecular Weight 98.08

10. Stability and reactivity

Reactive Hazard Yes

Stability Reacts violently with water. Hygroscopic.

Conditions to Avoid Incompatible products. Excess heat. Exposure to moist air or water. **Incompatible Materials** Water, Organic materials, Strong acids, Strong bases, Metals, Alcohols, Cyanides, Sulfides**Hazardous Decomposition Products** Sulfur oxides, Hydrogen

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information
Oral LD50 Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. **Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. **Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l. **Component Information**

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Sulfuric acid	2140 mg/kg (Rat)	Not listed	LC50 85 - 103 mg/m^3 (Rat) 1 h
Water	-	-	-

No information available

Toxicologically Synergistic Products
Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** Causes severe burns by all exposure routes **Sensitization** No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.Exposure to strong inorganic mists containing sulfuric acid may cause cancer byinhalation.

Component	CAS-No	IARC	NT	ACGIH	OSH	Mexi
Sulfuric acid	7664-93-9	Group 1	Known	A2	X	A2
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed

IARC (International Agency for Research on Cancer) *IARC (International Agency for Research on Cancer) Group 1 - Carcinogenic to Humans*

Group 2A - Probably Carcinogenic to Humans
Group 2B - Possibly Carcinogenic to Humans

NTP: (National Toxicity Program) *NTP: (National Toxicity Program) Known - Known Carcinogen*
Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen
A1 - Known Human Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)
A2 - Suspected Human Carcinogen A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

Mexico - Occupational Exposure Limits - Carcinogens *Mexico - Occupational Exposure Limits - Carcinogens A1 - Confirmed Human Carcinogen*
A2 - Suspected Human Carcinogen
A3 - Confirmed Animal Carcinogen
A4 - Not Classifiable as a Human Carcinogen
A5 - Not Suspected as a Human Carcinogen

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure Respiratory system
STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects,both acute and delayed
Possible perforation of stomach or esophagus should be investigated: Ingestioncausessevere swelling, severe damage to the delicate tissue and danger of perforation
Product is a corrosive material. Use of gastric lavage or emesis is contraindicated.

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity
This product contains the following substance(s) which are hazardous for the environment. .

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Sulfuric acid	-	LC50: > 500 mg/L, 96h static(Brachydanio rerio)	-	EC50: 29 mg/L/24h

Persistence and Degradability No information available

Bioaccumulation/ Accumulation No information available.

Mobility No information available.

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT
UN-No UN1830
Proper Shipping Name Sulfuric acid
Hazard Class 8
Packing Group II

TDG
UN-No UN1830
Proper Shipping Name SULFURIC ACID
Hazard Class 8
Packing Group II

IATA
UN-No UN1830
Proper Shipping Name SULFURIC ACID
Hazard Class 8
Packing Group II

IMDG/IMO
UN-No UN1830
Proper Shipping Name SULFURIC ACID
Hazard Class 8
Packing Group II

15. Regulatory information

United States of America Inventory

Sulfuric acid Revision
Date14-Feb-2020

Component	CAS-No	TSCA	TSCA Inventory notification - Active/Inactive	TSCA - EPA RegulatoryFlags
Sulfuric acid	7664-93-9	X	ACTIVE	-
Water	7732-18-5	X	ACTIVE	-

Legend:
TSCA - Toxic Substances Control Act, (40 CFR Part 710)
X - Listed
'-' - Not Listed

TSCA 12(b) - Notices of Export Not applicable

International Inventories
Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Australia (AICS), China (IECSC), Korea (ECL).

Component	CAS-No		DSL	NDSL	EINECS	PICC	EN	A	IECSCKECL
Sulfuric acid	7664-93-9		X	-231-639-5		X	X		X K
Water	7732-18-5		X	-231-791-2		X	X		X K

U.S. Federal Regulations

SARA 313

Component	CAS-No	Weight %	SARA 313 - Threshold Values %
Sulfuric acid	7664-93-9	90 - 98	1.0

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act)

Component	CWA - Hazardous Substances	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants
Sulfuric	X	1000 lb	-	-

Clean Air Act Not applicable

Not applicable

OSHA - Occupational Safety and Health Administration

CERCLA This material, as supplied, contains one or more substances regulated as a hazardoussubstance under the Comprehensive Environmental Response CompensationandLiability Act (CERCLA) (40 CFR 302)

Component	Hazardous Substances RQs	CERCLA EHS RQs
Sulfuric acid	1000 lb	1000 lb

California Proposition 65 This product contains the following Proposition 65 chemicals.

Component	CAS-No	California Prop. 65	Prop 65 NSRL	Category
Sulf	7664-93-9	Carcinogen	-	Carcinogen

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Sulf	X	X	X	X	X
Water	-	-	X	-	-

U.S. Department of Transportation
Reportable Quantity (RQ): Y
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

This product does not contain any DHS chemicals.

U.S. Department of Homeland Security

Other International Regulations

Mexico - Grade No information available

16. Other information

Prepared By Health, Safety and Environmental Department

Email: tech@alfa.com
www.alfa.com

Creation Date 12-Nov-2010
Revision Date 14-Feb-2020
Print Date 14-Feb-2020
Revision Summary SDS authoring systems update, replaces ChemGes SDS No. 7664-93-9/3.

Disclaimer
The information provided in this Safety Data Sheet is correct to the best of our knowledge, information andbelief at thedate of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,transportation, disposal and release and is not to be considered a warranty or quality specification. The informationrelates only to the specific material designated and may not be valid for such material used in combinationwithanyothermaterials or in any process, unless specified in the text

End of SDS

Safety Data Sheet (SDS)

SECTION 1 — CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Lead Sulfate

Flinn Scientific, Inc. P.O. Box 219, Batavia, IL 60510 (800) 452-1261
Chemtrec Emergency Phone Number: (800) 424-9919

SECTION 2 — HAZARDS IDENTIFICATION

Hazard class: Acute toxicity, oral and inhalation (Category 4). Harmful if swallowed or inhaled (H302+H332). Do not eat, drink or smoke when using this product (P270). Avoid breathing dust and fumes (P261).

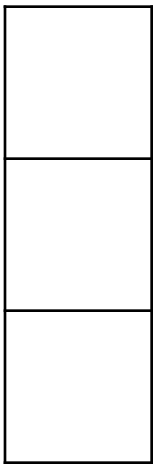
Hazard class: Carcinogenicity (Category 1B). May cause cancer (H350). Obtain special instructions before use (P201). Do not handle until all safety precautions have been read and understood (P202). Use personal protective equipment as required (P281). Inorganic lead compounds are probable human carcinogens (IARC 2A, NTP reasonably anticipated to be a human carcinogen).

Hazard class: Reproductive toxicity (Category 1A). May damage fertility or the unborn child (H360).

Hazard class: Specific target organ toxicity, repeated exposure (Category 2). May cause damage to organs through prolonged or repeated exposure (H373).

Signal Word

Pictograms



Component Name

Lead sulfate

Synonyms: Lead(II) sulfate

CAS Number	Formula
7446-14-2	$PbSO_4$

SECTION 3 — COMPOSITION, INFORMATION ON INGREDIENTS

DANGER

SECTION 4 — FIRST AID MEASURES

If exposed or concerned: Get medical advice or attention (P308+P313).
If inhaled: Remove victim to fresh air and keep at rest in a position comfortable for breathing (P304+P340). **If in eyes:** Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do so. Continue rinsing.
If on skin: Rinse with plenty of water.
If swallowed: Rinse mouth. Call a POISON CENTER or physician if you feel unwell

(P301+P330+P312). **SECTION 5 — FIRE FIGHTING MEASURES**

Nonflammable solid.
When heated to decomposition, may emit toxic fumes. In case of fire: Use a tri-class dry chemical fire extinguisher.

NFPA Code
None established

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Sweep up the spill, place in a sealed bag or container, and dispose. Ventilate area and wash spill site after material pickup is complete. See Sections 8 and 13 for further information.

SECTION 7 — HANDLING AND STORAGE

Flinn Suggested Chemical Storage Pattern: Inorganic #2. Store with acetates, halides, sulfates, sulfites, thiosulfates and phosphates.

Store in a cool, dry place. Store in a Flinn Chem-Saf™ bag. Use only in a hood or well-ventilated area (P271).

SECTION 8 — EXPOSURE CONTROLS, PERSONAL PROTECTION

Wear protective gloves, protective clothing, and eye protection. Wash hands thoroughly after handling (P264). Use only in a hood or well-ventilated area (P271).

Exposure guidelines: PEL/TLV 0.05 mg/m (OSHA/ACGIH) ^{3}

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

White rhombic crystals. Odorless.

Soluble: Sodium hydroxide. Slightly in hot water. Insoluble in alcohol.

strong bases and potassium. Shelf life: Indefinite, if stored properly.

Melting point: 1170 °C Specific gravity: 6.2

SECTION 10 — STABILITY AND REACTIVITY Avoid contact with

SECTION 11 — TOXICOLOGICAL INFORMATION

Acute effects: Convulsions, seizures, weakness, muscle cramps, methemoglobinemia.

Chronic effects: Anemia, reproductive hazard, probable carcinogen.

Target organs: Nerves, brain, blood, kidneys, female/male reproductive system

SECTION 12 — ECOLOGICAL INFORMATION Very toxic to aquatic life with long lasting effects.

SECTION 13 — DISPOSAL CONSIDERATIONS

ORL-GPIG LDL : 3000 mg/kg
0

IHL-RAT LC : N.A.
50

SKN-RBT LD : N.A.
50

Please review all federal, state and local regulations that may apply before proceeding.

Flinn Suggested Disposal Method #27f is one option.

SECTION 14 — TRANSPORT INFORMATION

Shipping name: Lead compounds, soluble, N.O.S. Hazard class: 6.1, Keep away from food. UN number: UN2291

SECTION 15 — REGULATORY INFORMATION

TSCA-listed, EINECS-listed (231-198-9), RCRA code D008.

SECTION 16 — OTHER INFORMATION

This Safety Data Sheet (SDS) is for guidance and is based upon information and tests believed to be reliable. Flinn Scientific, Inc. makes no guarantee of the accuracy or completeness of the data and shall not be liable for any damages relating thereto. The data is offered solely for your consideration, investigation, and verification. The data should not be confused with local, state, federal or insurance mandates, regulations, or requirements and CONSTITUTE NO WARRANTY. Any use of this data and information must be determined by the science instructor to be in accordance with applicable local, state or federal laws and regulations. The conditions or methods of handling, storage, use and disposal of the product(s) described are beyond the control of Flinn Scientific, Inc. and may be beyond our knowledge. FOR THIS AND OTHER REASONS, WE DO NOT ASSUME RESPONSIBILITY AND EXPRESSLY DISCLAIM LIABILITY FOR LOSS, DAMAGE OR EXPENSE ARISING OUT OF OR IN ANY WAY CONNECTED WITH THE HANDLING, STORAGE, USE OR DISPOSAL OF THIS

PRODUCT(S).

N.A. = Not available, not all health aspects of this substance have been fully investigated.

N/A = Not applicable

Consult your copy of the Flinn Science Catalog/Reference Manual for additional information about laboratory chemicals.

Revision Date: March 25, 2014

SAFETY DATA SHEET

Creation Date 11-Nov-2011 **Revision Date** 14-Feb-2020 **RevisionNumber**2

1. Identification

Product Name Lead plate

Cat No. : 12603

CAS-No 7439-92-1
Synonyms Lead metal

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.
Details of the supplier of the safety data sheet

Company
Alfa Aesar
Thermo Fisher Scientific Chemicals, Inc.
30 Bond Street
Ward Hill, MA 01835-8099
Tel: 800-343-0660
Fax: 800-322-4757
Email: tech@alfa.com
www.alfa.com

Emergency Telephone Number
During normal business hours (Monday-Friday, 8am-7pm EST), call (800) 343-0660.
After normal business hours, call Carechem 24 at (866) 928-0789.

2. Hazard(s) identification

Classification
This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Carcinogenicity Category 1B Reproductive Toxicity Category 1A Effects on or via lactation
Specific target organ toxicity - (repeated exposure) Category 1 Target Organs - Kidney, Central nervous system (CNS), Blood.

Label Elements

Signal Word
Danger

Hazard Statements
May cause cancer
May damage fertility. May damage the unborn child
May cause harm to breast-fed children
Causes damage to organs through prolonged or repeated exposure



Precautionary Statements
Prevention
Obtain special instructions before use
Do not handle until all safety precautions have been read and understood

Use personal protective equipment as required
Do not breathe dust/fume/gas/mist/vapors/spray
Avoid contact during pregnancy/while nursing
Wash face, hands and any exposed skin thoroughly after handling
Do not eat, drink or smoke when using this product

Response

IF exposed or concerned: Get medical attention/advice

Storage

Store locked up

Disposal

Dispose of contents/container to an approved waste disposal plant

Hazards not otherwise classified (HNOC)

WARNING. Cancer and Reproductive Harm - https://www.p65warnings.ca.gov/.

3. Composition/Information on Ingredients

Component	CAS-No	Weight %
Lead	7439-92-1	>95

4. First-aid measures

General Advice Show this safety data sheet to the doctor in attendance. Immediate medical attentionisrequired.

Eye Contact In the case of contact with eyes, rinse immediately with plenty of water and seekmedical advice.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Inhalation Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouthmethod if victim ingested or inhaled the substance; give artificial respiration withtheaidofa pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

Ingestion Do NOT induce vomiting. Call a physician or poison control center immediately. None reasonably foreseeable.

Most important symptoms and effects

Notes to Physician Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. **Unsuitable**

Extinguishing Media No information available

Flash Point No information available
Method - No information available

Autoignition Temperature No information available

Explosion Limits

Upper No data available
Lower No data available
Sensitivity to Mechanical Impact No information available

Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical
Non-combustible. Do not allow run-off from fire-fighting to enter drains or water courses.

Hazardous Combustion Products
Lead. lead oxides.
Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

NFPA

Health	Flammability	Instability	Physical hazards
2	0	0	N/A

6. Accidental release measures

Personal Precautions Ensure adequate ventilation. Use personal protective equipment as required. Avoiddust formation. Keep people away from and upwind of spill/leak. Evacuate personnel tosafe areas.

Environmental Precautions Do not flush into surface water or sanitary sewer system. Should not be releasedintotheenvironment. Do not allow material to contaminate ground water system.

Methods for Containment and Clean Up
Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

7. Handling and storage

Handling Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/faceprotection. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

Storage Keep in a dry, cool and well-ventilated place. Refer product specification and/or product label for specific storage temperature requirement. Keep container tightly closed.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Lead	TWA: 0.05 mg/m^3	TWA:	IDLH: 100 mg/m^3 TWA: 0.050 mg/m^3	TWA: 0.05 mg/m^3

Legend

ACGIH *- American Conference of Governmental Industrial Hygienists*
OSHA *- Occupational Safety and Health Administration*
NIOSH IDLH: *NIOSH - National Institute for Occupational Safety and Health*

Engineering Measures Ensure adequate ventilation, especially in confined areas. **Personal Protective**

Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166.

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or EuropeanStandardEN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State Solid
Appearance Grey
Odor Odorless
Odor Threshold No information available **pH** No information available **Melting Point/Range** 327.4 °C / 621.3 °F
Boiling Point/Range 1740 °C / 3164 °F @ 760 mmHg **Flash Point** No information available **Evaporation Rate** Not applicable
Flammability (solid,gas) No information available **Flammability or explosive limits**
Upper No data available
Lower No data available
Vapor Pressure 1.77 mmHg @ 1000 °C **Vapor Density** Not applicable **Specific Gravity** No information available
Solubility Insoluble in water
Partition coefficient; n-octanol/water No data available **Autoignition Temperature** No information available
Decomposition Temperature No information available **Viscosity** Not applicable **Molecular Formula** Pb
Molecular Weight 207.19

10. Stability and reactivity

Reactive Hazard None known, based on information available **Stability** Stable under normal conditions.

Conditions to Avoid Exposure to air. Incompatible products.

Incompatible Materials Strong acids, Ammonium nitrate: fertilizers capable of self-sustaining decomposition, Peroxides

Hazardous Decomposition Products Lead, lead oxides

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information

Component Information

No information available

Toxicologically Synergistic Products

Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** No information available

Sensitization May cause sensitization by skin contact

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.

Component	CAS-No	IARC	NT	ACGIH	OSH	Mexic
Lead	7439-92-1	Group 2A	Reasonab Anticipated	A3	X	A3

IARC (International Agency for Research on Cancer) *IARC (International Agency for Research on Cancer) Group 1 - Carcinogenic to Humans*
Group 2A - Probably Carcinogenic to Humans
Group 2B - Possibly Carcinogenic to Humans

NTP: (National Toxicity Program) *NTP: (National Toxicity Program) Known - Known Carcinogen*
Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists) *ACGIH: (American Conference of Governmental Industrial Hygienists)*
A1 - Known Human Carcinogen
A2 - Suspected Human Carcinogen A3 - Animal Carcinogen

Mexico - Occupational Exposure Limits - Carcinogens *Mexico - Occupational Exposure Limits - Carcinogens A1 - Confirmed Human Carcinogen*
A2 - Suspected Human Carcinogen
A3 - Confirmed Animal Carcinogen
A4 - Not Classifiable as a Human Carcinogen
A5 - Not Suspected as a Human Carcinogen

Mutagenic Effects No information available

Reproductive Effects May cause harm to the unborn child. Possible risk of impaired fertility. **Developmental Effects** No information available.

Teratogenicity No information available.

STOT - single exposure None known
STOT - repeated exposure Kidney Central nervous system (CNS) Blood **Aspiration hazard** No information available

No information available

Symptoms / effects,both acute and delayed

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

The product contains following substances which are hazardous for the environment. Contains a substance which is:. Verytoxictoaquatic organisms. May cause long-term adverse effects in the environment. Do not allow material to contaminate groundwatersystem.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea

Lead	Not listed	LC50: = 1.17 mg/L, 96h flow-through (Oncorhynchusmykiss) LC50: = 1.32 mg/L, 96hstatic (Oncorhynchus mykiss) LC50: = 0.44 mg/L, 96hsemi-static (Cyprinus carpio)	Not listed	EC50: = 600µ 48h(water flea)

Persistence and Degradability Insoluble in water May persist

Bioaccumulation/ Accumulation No information available.

Mobility Is not likely mobile in the environment due its low water solubility.

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT Not regulated
TDG Not regulated
IATA Not regulated
IMDG/IMO Not regulated

15. Regulatory information

United States of America Inventory

Component	CAS-No	TSCA	TSCA Inventory notification - Active/Inactive	TSCA - EPA RegulatoryFlags
Lead	7439-92-1	X	ACTIVE	-

Legend:
TSCA - Toxic Substances Control Act, (40 CFR Part 710)
Group 1 - Carcinogenic to Humans
Group 2A - Probably Carcinogenic to Humans
Group 2B - Possibly Carcinogenic to Humans

TSCA 12(b) - Notices of Export Not applicable

International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Australia (AICS), China (IECSC), Korea (ECL).

Component	CAS-No		DSL	NDSL	EINECS	PICC	EN	A	IECSCKECL
Lead	7439-92-1		X	- 231-100-4		X	X		X K

U.S. Federal Regulations

SARA 313

Component	CAS-No	Weight %	SARA 313 - Threshold Values %
Lead	7439-92-1	>95	0.1

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act)

Component	CWA - Hazardous	CWA - Reportable	CWA - Toxic Pollutants	CWA - Priority Pollutants

	Substances	Quantities		
Lead	-	-	X	X

Clean Air Act

Not applicable

OSHA - Occupational Safety and Health Administration

Component	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Lead	30 µg/m^3 Action Level 50 µg/m^3 TWA	-

CERCLA This material, as supplied, contains one or more substances regulated as a hazardoussubstance under the Comprehensive Environmental Response CompensationandLiability Act (CERCLA) (40 CFR 302)

Component	Hazardous Substances RQs	CERCLA EHS RQs
Lead	10 lb	-

California Proposition 65 This product contains the following Proposition 65 chemicals.

Component	CAS-No	California Prop. 65	Prop 65 NSRL	Category
Lead	7439-92-1	Carcinogen Developmental Female Reproductive Male Reproductive	15 µg/day	Developmental Carcinogen

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island

Lead	X	X	X	X	X

U.S. Department of Transportation
Reportable Quantity (RQ): N
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

U.S. Department of Homeland Security

This product does not contain any DHS chemicals.

Other International Regulations

Mexico - Grade No information available

16. Other information

Prepared By Health, Safety and Environmental Department

Email: tech@alfa.com
www.alfa.com

Creation Date 11-Nov-2011
Revision Date 14-Feb-2020
Print Date 14-Feb-2020
Revision Summary SDS authoring systems update, replaces ChemGes SDS No. 7439-92-1/3.

Disclaimer
The information provided in this Safety Data Sheet is correct to the best of our knowledge, information andbelief at thedate of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,transportation, disposal and release and is not to be considered a warranty or quality specification. The informationrelates only to the specific material designated and may not be valid for such material used in combinationwithanyothermaterials or in any process, unless specified in the text

End of SDS

Creation Date 22-Sep-2009 **Revision Date** 24-Dec-2021 **RevisionNumber**12

1. Identification

Product Name Hydrogen Peroxide (30% in water) Cat No. : BP2633-500; NC1592410

Synonyms Hydrogen Dioxide; Peroxide; Carbamide Peroxide

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company
Fisher Scientific Company
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11 Emergency Number
US:001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

2. Hazard(s) identification

Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Oxidizing liquids Category 2
Acute oral toxicity Category 4 Acute Inhalation Toxicity - Dusts and Mists Category 4 Skin Corrosion/Irritation Category 1 A Serious Eye Damage/Eye Irritation Category 1

Label Elements

Signal Word
Danger

Hazard Statements
May cause fire or explosion; strong oxidizer
Causes severe skin burns and eye damage
Harmful if swallowed or if inhaled



Precautionary Statements
Prevention
Wash face, hands and any exposed skin thoroughly after handling
Do not eat, drink or smoke when using this product
Use only outdoors or in a well-ventilated area
Do not breathe dust/fume/gas/mist/vapors/spray

Wear protective gloves/protective clothing/eye protection/face protection
Keep away from heat/sparks/open flames/hot surfaces. - No smoking
Keep/Store away from clothing/ other combustible materials
Take any precaution to avoid mixing with combustibles
Response
Immediately call a POISON CENTER or doctor/physician
Inhalation
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing **Skin**
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower Wash contaminated clothing before reuse
Eyes
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do.
Continuerinsing**Ingestion**
Rinse mouth
Do NOT induce vomiting
Fire
In case of fire: Use CO_2, dry chemical, or foam for extinction
Storage
Store locked up
Store in a well-ventilated place. Keep container tightly closed
Disposal
Dispose of contents/container to an approved waste disposal plant
Hazards not otherwise classified (HNOC)
None identified

3. Composition/Information on Ingredients

Component	CAS No	Weight %
Water	7732-18-5	65 - 80
Hydrogen peroxide	7722-84-1	20 - 35

4. First-aid measures

General Advice Show this safety data sheet to the doctor in attendance. Immediate medical attentionisrequired.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Remove andwashcontaminated clothing and gloves, including the inside, before re-use. Call a physician

immediately.

Inhalation If not breathing, give artificial respiration. Remove from exposure, lie down. Donot usemouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

Ingestion Do NOT induce vomiting. Clean mouth with water. Never give anything by mouthtoanunconscious person. Call a physician immediately.

Most important symptoms and effects

Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagusshouldbe investigated: Ingestion causes severe swelling, severe damage to the delicatetissueand danger of perforation

Notes to Physician Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media CO_2, dry chemical, dry sand, alcohol-resistant foam.

Unsuitable Extinguishing Media No information available

Flash Point Not applicable
Method - No information available

Autoignition Temperature No information available
Explosion Limits
Upper No data available
Lower No data available
Oxidizing Properties Oxidizer

Sensitivity to Mechanical Impact No information available
Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical
Corrosive material. The product causes burns of eyes, skin and mucous membranes. Containers may explode when heated. Oxidizer: Contact with combustible/organic material may cause fire. In the event of fire and/or explosion do not breathefumes. Thermal decomposition can lead to release of irritating gases and vapors. May ignite combustibles (wood paper, oil, clothing, etc.).

Hazardous Combustion Products
Hydrogen. Oxygen.
Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

NFPA

Health	Flammability	Instability	Physical hazards
3	0	2	OX

6. Accidental release measures

Personal Precautions Use personal protective equipment as required. Ensure adequate ventilation. Evacuatepersonnel to safe areas. Keep people away from and upwind of spill/leak.

Environmental Precautions Should not be released into the environment. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

7. Handling and storage

Handling Wear personal protective equipment/face protection. Do not get in eyes, on skin, or onclothing. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Ensure adequate ventilation.

Storage. Keep container tightly closed in a dry and well-ventilated place. Keep cool andprotect fromsunlight. Keep away from combustible material. Keep only in the original container. Donot store in metal containers. Corrosives area. Incompatible Materials. Strong oxidizing agents. copper. Finely powdered metals. Reducing Agent. Strong bases. Strongreducing agents. Combustible material.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Hydrogen peroxide	TWA: 1 ppm	(Vacated) TWA: 1 ppm (Vacated) TWA: 1.4 mg/m^{3}TWA: 1 ppm TWA: 1.4 mg/m^3	IDLH: 75 ppm TWA: 1 ppm TWA: 1.4 mg/m^3	TWA: 1 ppm

Legend

ACGIH *- American Conference of Governmental Industrial Hygienists*
OSHA *- Occupational Safety and Health Administration*
NIOSH IDLH: *NIOSH - National Institute for Occupational Safety and Health*

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstationlocation. Ensure adequate ventilation, especially in confined areas.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166.

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or EuropeanStandardEN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State Liquid
Appearance Clear
Odor pungent
Odor Threshold No information available **pH** 3.3 (30 %)
Melting Point/Range -33 °C / -27.4 °F **Boiling Point/Range** 108 °C / 226.4 °F @ 760 mmHg **Flash Point** Not applicable
Evaporation Rate >1.0 (Butyl Acetate = 1.0) **Flammability (solid,gas)** Not applicable
Flammability or explosive limits
Upper No data available
Lower No data available

Vapor Pressure 23 mmHg @ 30 °C

Vapor Density 1.10
Specific Gravity 1.11
Solubility Miscible with water
Partition coefficient; n-octanol/water No data available **Autoignition Temperature** No information available
Decomposition Temperature No information available **Viscosity** No information available **Molecular Formula** H2O2
Molecular Weight 34.01

10. Stability and reactivity

Reactive Hazard Yes

Stability Oxidizer: Contact with combustible/organic material may cause fire. Light sensitive. **Conditions to Avoid** Incompatible products. Excess heat. Exposure to light. Combustible material.

Incompatible Materials Strong oxidizing agents, copper, Finely powdered metals, Reducing Agent, Strongbases,Strong reducing agents, Combustible material

Hazardous Decomposition Products Hydrogen, Oxygen

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information
Oral LD50 Category 4. ATE = 300 - 2000 mg/kg.
Dermal LD50 Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. **Mist LC50** Category 4. ATE = 1 - 5 mg/l.
Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Hydrogen peroxide	376 mg/kg (Rat) (90%) 910 mg/kg (Rat) (20-60%) 1518 mg/kg (Rat) (8-20% sol)	>2000 mg/kg (Rabbit)	LC50 = 2000 mg/m^3 (Rat) 4 h

No information available

Toxicologically Synergistic Products
Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** Causes burns by all exposure routes

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.

Component	CAS No	IARC	NT	ACGIH	OSH	Mexi

Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed
Hydrogen peroxide	7722-84-1	Not listed	Not listed	A3	Not listed	A3

IARC (International Agency for Research on Cancer) *IARC (International Agency for Research on Cancer) Group 1 - Carcinogenic to Humans*
Group 2A - Probably Carcinogenic to Humans
Group 2B - Possibly Carcinogenic to Humans
A1 - Known Human Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)
A2 - Suspected Human Carcinogen A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

Mexico - Occupational Exposure Limits - Carcinogens *Mexico - Occupational Exposure Limits - Carcinogens A1 - Confirmed Human Carcinogen*
A2 - Suspected Human Carcinogen
A3 - Confirmed Animal Carcinogen
A4 - Not Classifiable as a Human Carcinogen
A5 - Not Suspected as a Human Carcinogen

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known
STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects,both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestioncausessevere swelling, severe damage to the delicate tissue and danger of perforation

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity
Contains a substance which is:. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Do not empty into drains. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminategroundwater system.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Hydrogen peroxide	EC50 2.5 mg/L/72h	LC50: 16.4 mg/L/96h (P.promelas)	Not listed	EC50 7.7 mg/L/24h

Persistence and Degradability Persistence is unlikely based on information available. Miscible with water **Bioaccumulation/ Accumulation** No information available.

Mobility Will likely be mobile in the environment due to its water solubility.

Component	log Pow

Hydrogen peroxide	-1.1

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT

UN-No UN2014
Proper Shipping Name HYDROGEN PEROXIDE, AQUEOUS SOLUTIONS **Hazard Class** 5.1
Subsidiary Hazard Class 8
Packing Group II

TDG

UN-No UN2014
Proper Shipping Name HYDROGEN PEROXIDE, AQUEOUS SOLUTION

Hazard Class 5.1
Subsidiary Hazard Class 8
Packing Group II

IATA

UN-No UN2014
Proper Shipping Name HYDROGEN PEROXIDE, AQUEOUS SOLUTION **Hazard Class** 5.1
Subsidiary Hazard Class 8
Packing Group II

IMDG/IMO

UN-No UN2014
Proper Shipping Name HYDROGEN PEROXIDE, AQUEOUS SOLUTION **Hazard Class** 5.1
Subsidiary Hazard Class 8
Packing Group II

15. Regulatory information

United States of America Inventory

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	TSCA - EPA RegulatoryFlags
Water	7732-18-5	X	ACTIVE	-
Hydrogen peroxide	7722-84-1	X	ACTIVE	-

Legend:
TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)
X - Listed
'-' - Not Listed

TSCA 12(b) - Notices of Export Not applicable

International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea(KECL).

Component	CAS No		DSL	NDSL	EINECS	PI	E	I	AIC	IECSCKEC
Water	7732-18-5		X	231-791-						XKE-35400
Hydrogen peroxide	7722-84-1		X	231-765-						XKE-20204

KECL - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

U.S. Federal Regulations

SARA 313 Not applicable

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act) Not applicable

Clean Air Act Not applicable

OSHA - Occupational Safety and Health Administration

Component	Specifically Regulated Chemicals	Highly Hazardous Chemicals
Hydrogen peroxide	-	TQ: 7500 lb

CERCLA This material, as supplied, contains one or more substances regulated as a hazardoussubstance under the Comprehensive Environmental Response CompensationandLiability Act (CERCLA) (40 CFR 302)

Component	Hazardous Substances RQs	CERCLA EHS RQs
Hydrogen peroxide	-	1000 lb

California Proposition 65 This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Water	-	-	X	-	-
Hydrogen peroxide	X	X	X	-	X

U.S. Department of Transportation
Reportable Quantity (RQ): N
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

This product contains the following DHS chemicals:

U.S. Department of Homeland Security

placarded amount

Legend - STQs = Screening Threshold Quantities, APA = A

Component	DHS Chemical Facility Anti-TerrorismStandard
Hydrogen peroxide	Theft STQs - 400lb (concentration >=35

Other International Regulations

Mexico - Grade No information available

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV -Substances Subject to Authorization	REACH (1907/2006) - Annex XVII -Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Hydrogen peroxide	-	Use restricted. See item 75. (see link for restriction details)	-

https://echa.europa.eu/substances-restricted-under-reach

Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone DepletionPotential	Restriction of Hazardous Substances (RoHS)
Water	7732-18-5	Listed	Not applicable	Not applicable	Not applicable
Hydrogen peroxide	7722-84-1	Listed	Not applicable	Not applicable	Not applicable

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantitiesfor Major AccidentNotification	Seveso III Directive(2012/18/EC) - Qualifying Quantitiesfor Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention(Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
Hydrogen peroxide	7722-84-1	Not applicable	Not applicable	Not applicable	Not applicable

16. Other information

Prepared By Regulatory Affairs

Thermo Fisher Scientific

Email: EMSDS.RA@thermofisher.com

Creation Date 22-Sep-2009

Revision Date 24-Dec-2021
Print Date 24-Dec-2021
Revision Summary SDS sections updated. 7.

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information andbelief at thedate of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,transportation, disposal and release and is not to be considered a warranty or quality specification. The informationrelates only to the specific material designated and may not be valid for such material used in combinationwithanyothermaterials or in any process, unless specified in the text

End of SDS



SAFETY DATA SHEET

Creation Date 05-May-2009 **Revision Date** 24-Dec-2021 **RevisionNumber**6

1. Identification

Product Name Acetic acid

Cat No. : A35-500; A38-212; A38-450LB; A38-500; A38-500LC; A38C-212; A38C-212EA; A38P-20; A38P-500; A38S-212; A38S-500; A38SI-212; A465-1; A465-250; A465-500; A490-212; A490-212LC; A491-212; BP1185-500; BP1185-500LC; BP2400-500; BP2401-212; BP2401-500; BP2401C-212; BP2401P-20; BP2401S-212; BP2401S-500;

CAS No 64-19-7
Synonyms Glacial acetic acid; Methanecarboxylic acid; Ethanoic acid; Vinegar acid (HPLC/CertifiedACS/OPTIMA//USP/FCC/EP/BP/Trace Metal Grade/Aldehyde-Free/Sequencing)

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company
Fisher Scientific Company
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number CHEMTRECÒ, Inside the USA: 800-424-9300 CHEMTRECÒ, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification
This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable liquids Category 3
Skin Corrosion/Irritation Category 1 A Serious Eye Damage/Eye Irritation Category 1

Label Elements

Signal Word
Danger

Hazard Statements
Flammable liquid and vapor
Causes severe skin burns and eye damage




Precautionary Statements
Prevention
Keep away from heat/sparks/open flames/hot surfaces. - No smoking
Take precautionary measures against static discharge
Wear protective gloves/protective clothing/eye protection/face protection
Do not breathe dust/fume/gas/mist/vapors/spray
Use only outdoors or in a well-ventilated area
Wash face, hands and any exposed skin thoroughly after handling
Keep container tightly closed
Response

Immediately call a POISON CENTER or doctor/physician
Inhalation
IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing **Skin**
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower Wash contaminated clothing before reuse
Eyes
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continuerinsing**Ingestion**
IF SWALLOWED: Rinse mouth. DO NOT induce vomiting
Fire
In case of fire: Use CO2, dry chemical, or foam for extinction
Storage
Store locked up
Store in a well-ventilated place. Keep container tightly closed
Disposal
Dispose of contents/container to an approved waste disposal plant
Hazards not otherwise classified (HNOC)
None identified

3. Composition/Information on Ingredients

Component	CAS No	Weight %
Acetic acid	64-19-7	>95

4. First-aid measures

General Advice Show this safety data sheet to the doctor in attendance. Immediate medical attentionisrequired.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Remove andwashcontaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

Inhalation If not breathing, give artificial respiration. Remove from exposure, lie down. Donot usemouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

Ingestion Do NOT induce vomiting. Clean mouth with water. Never give anything by mouthtoanunconscious person. Call a physician immediately.

Most important symptoms and effects Causes burns by all exposure routes. Ingestion causes severe swelling, severedamageto the delicate tissue and danger of perforation: Symptoms of overexposure may beheadache, dizziness, tiredness, nausea and vomiting

Notes to Physician Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media CO_2, dry chemical, dry sand, alcohol-resistant foam. **Unsuitable**

Extinguishing Media No information available

Flash Point 40 °C / 104 °F

Method - No information available

Autoignition Temperature 427 °C / 800.6 °F

Explosion Limits
Upper 19.9 vol %
Lower 4.0 vol %
Sensitivity to Mechanical Impact No information available
Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical
Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin andmucousmembranes.

Hazardous Combustion Products
Carbon monoxide (CO). Carbon dioxide (CO_2). Thermal decomposition can lead to release of irritating gases and vapors. **Protective Equipment and Precautions for Firefighters**
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

NFPA

Health	Flammability	Instability	Physical hazards
3	2	0	N/A

6. Accidental release measures

Personal Precautions Use personal protective equipment as required. Ensure adequate ventilation. Evacuatepersonnel to safe areas. Keep people away from and upwind of spill/leak.

Environmental Precautions Should not be released into the environment.

Methods for Containment and Clean Up Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

7. Handling and storage

Handling Do not get in eyes, on skin, or on clothing. Wear personal protective equipment/faceprotection. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Donot ingest. If swallowed then seek immediate medical assistance.

Storage. Corrosives area. Keep away from heat, sparks and flame. Keep containers tightlyclosedina dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizingagents. Strong bases. Metals.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Acetic acid	TWA: 10 ppm STEL: 15 ppm	(Vacated) TWA: 10 ppm(Vacated) TWA: 25 mg/m^3 TWA: 10 ppm TWA: 25 mg/m^3	IDLH: 50 ppm TWA: 10 ppm TWA: 25 mg/m^3 STEL: 15 ppm STEL: 37 mg/m^3	TWA: 10 ppm STEL: 15 ppm

Legend

ACGIH *- American Conference of Governmental Industrial Hygienists*
OSHA *- Occupational Safety and Health Administration*
NIOSH IDLH: *NIOSH - National Institute for Occupational Safety and Health*

Engineering Measures Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lightingequipment. Ensure that eyewash stations and safety showers are close to theworkstation location. Ensure adequate ventilation, especially in confined areas.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166. Tight sealing safety goggles. Face protection shield.

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or EuropeanStandardEN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State Liquid
Appearance Colorless
Odor vinegar-like
Odor Threshold No information available **pH** < 2.5 10 g/L aq.sol
Melting Point/Range 16 - 16.5 °C / 60.8 - 61.7 °F **Boiling Point/Range** 117 - 118 °C / 242.6 - 244.4 °F **Flash Point** 40 °C / 104 °F **Evaporation Rate** 0.97 (Butyl Acetate = 1.0) **Flammability (solid,gas)** Not applicable
Flammability or explosive limits
Upper 19.9 vol % **Lower** 4.0 vol % **Vapor Pressure** 1.52 kPa @ 20 °C **Vapor Density** 2.10
Specific Gravity 1.048

Solubility Soluble in water
Partition coefficient; n-octanol/water No data available **Autoignition Temperature** 427 °C / 800.6 °F **Decomposition Temperature** No information available **Viscosity** 1.53 mPa.s @ 25 °C **Molecular Formula** C2 H4 O2
Molecular Weight 60.05

10. Stability and reactivity

Reactive Hazard None known, based on information available **Stability** Stable under normal conditions.

Conditions to Avoid Incompatible products. Excess heat. Keep away from open flames, hot surfacesandsources of ignition.

Incompatible Materials Strong oxidizing agents, Strong bases, Metals

Hazardous Decomposition Products Carbon monoxide (CO), Carbon dioxide (CO_2), Thermal decomposition can leadtoreleaseof irritating gases and vapors

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information
Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Acetic acid	3310 mg/kg (Rat)	-	> 40 mg/L (Rat) 4 h

No information available

Toxicologically Synergistic Products
Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** Causes severe burns by all exposure routes **Sensitization** No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.

Component	CAS No	IARC	NT	ACGIH	OSH	Mexi
Acetic acid	64-19-7	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects Not mutagenic in AMES Test

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known
STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects,both acute and delayed Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

Do not empty into drains.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Acetic acid	-	Pimephales promelas: LC50= 88 mg/L/96h Lepomis macrochirus: LC50= 75 mg/L/96h	Photobacterium phosphoreum: EC50 = 8.8 mg/L/15 min Photobacterium phosphoreum: EC50 = 8.8mg/L/25 min Photobacterium phosphoreum: EC50 = 8.8mg/L/5 min	EC50 = 95 mg/L/24h

Persistence and Degradability Miscible with water Persistence is unlikely based on information available. **Bioaccumulation/**

Accumulation No information available.

Mobility Will likely be mobile in the environment due to its water solubility.

Component	log Pow
Acetic acid	-0.2

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT

UN-No UN2789
Proper Shipping Name ACETIC ACID, GLACIAL
Hazard Class 8
Subsidiary Hazard Class 3
Packing Group II

TDG

UN-No UN2789
Proper Shipping Name ACETIC ACID, GLACIAL
Hazard Class 8
Subsidiary Hazard Class 3
Packing Group II

IATA

UN-No UN2789
Proper Shipping Name ACETIC ACID, GLACIAL
Hazard Class 8
Subsidiary Hazard Class 3

Packing Group II

IMDG/IMO

UN-No UN2789
Proper Shipping Name ACETIC ACID, GLACIAL
Hazard Class 8
Subsidiary Hazard Class 3
Packing Group II

15. Regulatory information

United States of America Inventory

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	TSCA - EPA RegulatoryFlags
Acetic acid	64-19-7	X	ACTIVE	-

Legend:
TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)
X - Listed
'-' - Not Listed

TSCA 12(b) - Notices of Export Not applicable

International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea(KECL).

Component	CAS No		DSL	NDSL	EINECS	PI	E	I	AIC	IECSCKEC
Acetic acid	64-19-7		X	200-580-7						XX

KECL - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

U.S. Federal Regulations

SARA 313 Not applicable

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act)

Component	CWA - Hazardous Substances	CWA - Reportable Quantities	CWA - Toxic Pollutants	CWA - Priority Pollutants
Acetic acid	X	5000 lb	-	-

Clean Air Act Not applicable

Not applicable

OSHA - Occupational Safety and Health Administration

CERCLA This material, as supplied, contains one or more substances regulated as a hazardoussubstance under the Comprehensive Environmental Response CompensationandLiability Act (CERCLA) (40 CFR 302)

Component	Hazardous Substances RQs	CERCLA EHS RQs
Acetic acid	5000 lb	-

California Proposition 65 This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Acetic acid	X	X	X	-	X

U.S. Department of Transportation
Reportable Quantity (RQ): Y
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

This product does not contain any DHS chemicals.

U.S. Department of Homeland Security

Other International Regulations

Mexico - Grade Moderate risk, Grade 2

Authorisation/Restrictions according to EU REACH

Component	REACH (1907/2006) - Annex XIV -Substances Subject to Authorization	REACH (1907/2006) - Annex XVII -Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Acetic acid	-	Use restricted. See item 75. (see link for restriction details)	-

https://echa.europa.eu/substances-restricted-under-reach

Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone DepletionPotential	Restriction of Hazardous Substances (RoHS)

Acetic acid	64-19-7	Listed	Not applicable	Not applicable	Not applicable

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantitiesfor Major AccidentNotification	Seveso III Directive(2012/18/EC) - Qualifying Quantitiesfor Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention(Hazardous Waste)
Acetic acid	64-19-7	Not applicable	Not applicable	Not applicable	Annex I - Y34

16. Other information

Prepared By Regulatory Affairs

Thermo Fisher Scientific

Email: EMSDS.RA@thermofisher.com

Creation Date 05-May-2009
Revision Date 24-Dec-2021
Print Date 24-Dec-2021

Revision Summary This document has been updated to comply with the US OSHA HazCom2012Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information andbelief at thedate of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,transportation, disposal and release and is not to be considered a warranty or quality specification. The informationrelates only to the specific material designated and may not be valid for such material used in combinationwithanyothermaterials or in any process, unless specified in the text

End of SDS

SAFETY DATA SHEET

Revision Date 25-Dec-2021 **RevisionNumber**4

1. Identification

Product Name Indium(III) iodide

CAS No 13510-35-5
Synonyms No information available

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company
Fisher Scientific Company
One Reagent Lane Fair Lawn, NJ 07410
Tel: (201) 796-7100

Acros Organics
One Reagent Lane Fair Lawn, NJ 07410

Emergency Telephone Number For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11 Emergency Number
US:001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

2. Hazard(s) identification

Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Skin Corrosion/Irritation Category 2 Serious Eye Damage/Eye Irritation Category 2 Respiratory Sensitization Category 1 Skin Sensitization Category 1 Reproductive Toxicity Category 1B
Specific target organ toxicity (single exposure) Category 3 Target Organs - Respiratory system.

Label Elements

Signal Word
Danger

Hazard Statements
Causes skin irritation
Causes serious eye irritation
May cause an allergic skin reaction
May cause allergy or asthma symptoms or breathing difficulties if inhaled
May cause respiratory irritation
May damage fertility or the unborn child




Precautionary Statements

Prevention
Obtain special instructions before use
Do not handle until all safety precautions have been read and understood
Use personal protective equipment as required
Wash face, hands and any exposed skin thoroughly after handling
Avoid breathing dust/fume/gas/mist/vapors/spray
In case of inadequate ventilation wear respiratory protection
Contaminated work clothing should not be allowed out of the workplace
Use only outdoors or in a well-ventilated area
Wear protective gloves/protective clothing/eye protection/face protection
Response
IF exposed or concerned: Get medical attention/advice
Inhalation
If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing **Skin**
IF ON SKIN: Wash with plenty of soap and water
Take off contaminated clothing and wash before reuse
If skin irritation or rash occurs: Get medical advice/attention
Eyes
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. ContinuerinsingIf eye irritation persists: Get medical advice/attention
Storage
Store locked up
Store in a well-ventilated place. Keep container tightly closed
Disposal
Dispose of contents/container to an approved waste disposal plant
Hazards not otherwise classified (HNOC)
None identified

3. Composition/Information on Ingredients

Component	CAS No	Weight %
Indium iodide	13510-35-5	99.99

4. First-aid measures

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Getmedical attention.

Skin Contact Wash off immediately with soap and plenty of water while removing all contaminatedclothes and shoes. Get medical attention.

Inhalation Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.

Ingestion Clean mouth with water. Get medical attention.

Most important symptoms and effects
May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing
Notes to Physician Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media Water spray. Carbon dioxide (CO_2). Dry chemical. Chemical foam. **Unsuitable**

Extinguishing Media No information available

Flash Point No information available
Method - No information available

Autoignition Temperature No information available
Explosion Limits
Upper No data available
Lower No data available
Sensitivity to Mechanical Impact No information available
Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical
Keep product and empty container away from heat and sources of ignition.

Hazardous Combustion Products
Hydrogen iodide.
Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear.

NFPA

Health	Flammability	Instability	Physical hazards
2	0	1	N/A

6. Accidental release measures

Personal Precautions Ensure adequate ventilation. Use personal protective equipment as required. **Environmental Precautions** See Section 12 for additional Ecological Information.

Methods for Containment and Clean Up Sweep up and shovel into suitable containers for disposal. Do not let this chemical enterthe environment.

7. Handling and storage

Handling Avoid contact with skin and eyes. Do not breathe dust. Do not ingest. If swallowedthenseek immediate medical assistance.

Storage. Keep in a dry, cool and well-ventilated place. Keep container tightly closed. IncompatibleMaterials. Strong oxidizing agents. Finely powdered metals.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Indium iodide (InI3)	TWA: 0.1 mg/m^3	(Vacated) TWA: 0.1 mg/m^3	TWA: 0.1 mg/m^3	TWA: 0.1 mg/m^3

Legend

ACGIH *- American Conference of Governmental Industrial Hygienists*
OSHA *- Occupational Safety and Health Administration*
NIOSH IDLH: *NIOSH - National Institute for Occupational Safety and Health*

Engineering Measures Ensure adequate ventilation, especially in confined areas. Ensure that eyewashstationsand safety showers are close to the workstation location.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166.

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or EuropeanStandardEN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State Solid
Appearance Yellow-orange
Odor No information available **Odor Threshold** No information available **pH** No information available **Melting Point/Range** 210 °C / 410 °F
Boiling Point/Range No information available **Flash Point** No information available **Evaporation Rate** Not applicable
Flammability (solid,gas) No information available **Flammability or explosive limits**
Upper No data available
Lower No data available
Vapor Pressure No information available **Vapor Density** Not applicable
Specific Gravity No information available **Solubility** No information available **Partition coefficient; n-octanol/water** No data available **Autoignition Temperature** No information available **Decomposition Temperature** No information available **Viscosity** Not applicable
Molecular Formula In I3
Molecular Weight 495.53

10. Stability and reactivity

Reactive Hazard None known, based on information available

Stability Hygroscopic.

Conditions to Avoid Incompatible products. Exposure to moist air or water. **Incompatible Materials** Strong oxidizing agents, Finely powdered metals **Hazardous Decomposition Products** Hydrogen iodide

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information No acute toxicity information is available for this product **Component Information**

No information available

Toxicologically Synergistic Products

Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** No information available

Sensitization May cause sensitization by skin contact

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.

Component	CAS No	IARC	NT	ACGIH	OSH	Mexi
Indium iodide (InI3)	13510-35-5	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure Respiratory system
STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects,both acute and delayed Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity
Do not empty into drains.

Persistence and Degradability Soluble in water Persistence is unlikely based on information available. **Bioaccumulation/ Accumulation** No information available.

Mobility Will likely be mobile in the environment due to its water solubility.

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT Not regulated
TDG Not regulated
IATA Not regulated
IMDG/IMO Not regulated

15. Regulatory information

United States of America Inventory

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	TSCA - EPA RegulatoryFlags
Indium iodide (InI3)	13510-35-5	X	ACTIVE	-

Legend:
TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)
X - Listed
'-' - Not Listed

TSCA 12(b) - Notices of Export Not applicable

International Inventories
Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea(KECL).

Component	CAS No		DSL	NDSL	EINECS	PI	E	I	AIC	IECSCKEC
Indium iodide (InI3)	13510-35-5		- X			-				- XKE-21005

KECL - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

U.S. Federal Regulations

SARA 313 Not applicable

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act) Not applicable

Clean Air Act Not applicable

Not applicable

OSHA - Occupational Safety and Health Administration

CERCLA Not applicable

California Proposition 65 This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Component	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Indium	iodide (InI3) -	-	-	-	X

U.S. Department of Transportation
Reportable Quantity (RQ): N
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

U.S. Department of Homeland Security

This product does not contain any DHS chemicals.

Other International Regulations

Mexico - Grade No information available

Authorisation/Restrictions according to EU REACH

Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone DepletionPotential	Restriction of Hazardous Substances (RoHS)
Indium iodide (InI3)	13510-35-5	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantitiesfor Major AccidentNotification	Seveso III Directive(2012/18/EC) - Qualifying Quantitiesfor Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention(Hazardous Waste)
Indium iodide (InI3)	13510-35-5	Not applicable	Not applicable	Not applicable	Not applicable

16. Other information

Prepared By Regulatory Affairs

Thermo Fisher Scientific

Email: EMSDS.RA@thermofisher.com

Revision Summary This document has been updated to comply with the US OSHA HazCom2012Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

Disclaimer

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End of SDS



SAFETY DATA SHEET

Creation Date 14-Sep-2009 **Revision Date** 29-Oct-2018 **RevisionNumber**4

1. Identification

Product Name Potassium iodide

Cat No. : P410-10; P410-100; P410-3; P410-500

CAS-No 7681-11-0
Synonyms Knollide; Potide

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use **Details of the supplier of the safety data sheet**

Company
Fisher Scientific
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number
CHEMTRECÒ, Inside the USA: 800-424-9300
CHEMTRECÒ, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification
This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Specific target organ toxicity - (repeated exposure) Category 1

Label Elements

Signal Word
Danger

Hazard Statements
Causes damage to organs through prolonged or repeated exposure



Precautionary Statements
Prevention
Do not breathe dust/fume/gas/mist/vapors/spray
Wash face, hands and any exposed skin thoroughly after handling
Do not eat, drink or smoke when using this product
Response
Get medical attention/advice if you feel unwell
Disposal
Dispose of contents/container to an approved waste disposal plant
Hazards not otherwise classified (HNOC)
None identified

3. Composition/Information on Ingredients

Component	CAS-No	Weight %
Potass	7681-11-0	>95

4. First-aid measures

General Advice If symptoms persist, call a physician.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Getmedical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. If skin irritationpersists,call a physician.

Inhalation Move to fresh air. If not breathing, give artificial respiration. Get medical attentionif symptoms occur.

Ingestion Clean mouth with water and drink afterwards plenty of water. Get medical attentionif symptoms occur.

. May cause pulmonary edema

Most important symptoms and

effects
Notes to Physician Treat symptomatically

5. Fire-fighting measures

Unsuitable Extinguishing Media No information available

Flash Point No information available
Method - No information available

Autoignition Temperature
Explosion Limits
Upper No data available
Lower No data available
Sensitivity to Mechanical Impact No information available
Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical
Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to releaseof irritatinggases and vapors.

Hazardous Combustion Products
Hydrogen iodide
Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear.

NFPA

Health	Flammability	Instability	Physical hazards
1	0	1	N/A

6. Accidental release measures

Personal Precautions Ensure adequate ventilation. Use personal protective equipment. Avoid dust formation. **Environmental Precautions** Should not be released into the environment.

Methods for Containment and Clean Up Sweep up or vacuum up spillage and collect in suitable container for disposal. Keepin suitable, closed containers for disposal.

7. Handling and storage

Handling Wear personal protective equipment. Ensure adequate ventilation. Avoid ingestionandinhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

Storage Keep containers tightly closed in a dry, cool and well-ventilated place. Keep awayfromdirect sunlight. Store under an inert atmosphere. Protect from moisture.

8. Exposure controls / personal protection

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH IDLH	Mexico OEL (TWA)
Potassium iodide	TWA: 0.01 ppm			

Legend

ACGIH *- American Conference of Governmental Industrial Hygienists*

Engineering Measures Ensure adequate ventilation, especially in confined areas. Ensure that eyewashstationsand safety showers are close to the workstation location.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166.

Skin and body protection Long sleeved clothing.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or EuropeanStandardEN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State Solid
Appearance White
Odor Odorless
Odor Threshold No information available **pH** 6-8 5% in water (20°C) **Melting Point/Range** 680 °C / 1256 °F **Boiling Point/Range** 1330 °C / 2426 °F @ 760 mmHg **Flash Point** No information available **Evaporation Rate** Not applicable

Flammability (solid,gas) No information available **Flammability or explosive limits**
Upper No data available
Lower No data available
Vapor Pressure 1 mmHg @ 745 °C **Vapor Density** Not applicable
Specific Gravity No information available **Solubility** Soluble in water
Partition coefficient; n-octanol/water No data available **Autoignition Temperature**
Decomposition Temperature No information available **Viscosity** Not applicable
Molecular Formula I K
Molecular Weight 166

10. Stability and reactivity

Reactive Hazard None known, based on information available **Stability** Air sensitive. Light sensitive.

Hygroscopic.

Conditions to Avoid Excess heat. Avoid dust formation. Exposure to moist air or water. Exposure toair. Exposure to light.

Incompatible Materials Strong oxidizing agents

Hazardous Decomposition Products Hydrogen iodide

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information
Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Potassium iodide	2779 mg/kg (Rat)	Not listed	Not listed

No information available

Toxicologically Synergistic Products
Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** May cause irritation

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.

Component	CAS-No	IARC	NT	ACGIH	OSH	Mexi
Potassium iodide	7681-11-0	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known

STOT - repeated exposure None known

Aspiration hazard No information available

May cause pulmonary edema

Symptoms / effects,both acute and delayed

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

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Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea

Potassium iodide	-	Onchorhynchus mykiss:LC50: 3200 mg/L/120h	-	-

Persistence and Degradability Persistence is unlikely

Bioaccumulation/ Accumulation No information available.

Mobility . Will likely be mobile in the environment due to its water solubility.

Component	log Pow
Potassium iodide	0.04

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT Not regulated
TDG Not regulated
IATA Not regulated
IMDG/IMO Not regulated

15. Regulatory information

All of the components in the product are on the following Inventory lists: China X = listed Australia U.S.A. (TSCA) Canada(DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (ECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

International Inventories

Component	TSCA			DSL ND	EINECS	NLP					IECS
Po				- 231-6	-		X				

Legend:
X - Listed
E - Indicates a substance that is the subject of a Section 5(e) Consent order under TSCA.
F - Indicates a substance that is the subject of a Section 5(f) Rule under TSCA.
N - Indicates a polymeric substance containing no free-radical initiator in its inventory name but is considered to cover the designatedpolymer made with any free-radical initiator regardless of the amount used.
P - Indicates a commenced PMN substance
R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA.
S - Indicates a substance that is identified in a proposed or final Significant New Use Rule T - Indicates a substance that is the subject of a Section 4 test rule under TSCA.
XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data BaseProduction and Site Reports (40 CFR 710(B).
Y1 - Indicates an exempt polymer that has a number-average molecular weight of 1,000 or greater.

Y2 - Indicates an exempt polymer that is a polyester and is made only from reactants included in a specified list of low concern reactantsthat comprises one of the eligibility criteria for the exemption rule.

U.S. Federal Regulations

TSCA 12(b) Not applicable

SARA 313 Not applicable

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act) Not applicable

Clean Air Act Not applicable

OSHA Occupational Safety and Health Administration
Not applicable

CERCLA Not applicable

California Proposition 65 This product does not contain any Proposition 65 chemicals Not applicable

U.S. State Right-to-Know Regulations

U.S. Department of Transportation

Reportable Quantity (RQ): N
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

U.S. Department of Homeland Security
This product does not contain any DHS chemicals.

Other International Regulations

Mexico - Grade No information available

16. Other information

Prepared By Regulatory Affairs
Thermo Fisher Scientific
Email: EMSDS.RA@thermofisher.com

Creation Date 14-Sep-2009
Revision Date 29-Oct-2018
Print Date 29-Oct-2018
Revision Summary This document has been updated to comply with the US OSHA HazCom2012Standardreplacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

Disclaimer
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End of SDS

SAFETY DATA SHEET

Creation Date 20-Jul-2009 **Revision Date** 14-Feb-2020 **RevisionNumber**3

1. Identification

Product Name Sodium acetate, anhydrous Cat No. : A13184

CAS-No 127-09-3
Synonyms Sodium acetate

Recommended Use Laboratory chemicals.
Uses advised against Food, drug, pesticide or biocidal product use.
Details of the supplier of the safety data sheet

Company
Alfa Aesar
Thermo Fisher Scientific Chemicals, Inc.
30 Bond Street
Ward Hill, MA 01835-8099
Tel: 800-343-0660
Fax: 800-322-4757
Email: tech@alfa.com
www.alfa.com

Emergency Telephone Number
During normal business hours (Monday-Friday, 8am-7pm EST), call (800) 343-0660.
After normal business hours, call Carechem 24 at (866) 928-0789.

2. Hazard(s) identification

Classification
This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Combustible dust Yes

Label Elements

Signal Word
Warning

Hazard Statements
May form combustible dust concentrations in air

Precautionary Statements
Storage
Store in a well-ventilated place. Keep container tightly closed
Hazards not otherwise classified (HNOC)

None identified

3. Composition/Information on Ingredients

Component	CAS-No	Weight %
Sodium	127-09-3	>95

4. First-aid measures

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Getmedical attention.

Skin Contact Get medical attention immediately if symptoms occur. Wash off immediately withplentyofwater for at least 15 minutes.

Inhalation Remove to fresh air. Get medical attention immediately if symptoms occur.

Ingestion Clean mouth with water and drink afterwards plenty of water. Get medical attentionif symptoms occur.

None reasonably foreseeable.

Most important symptoms and effects

Notes to Physician Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media Water spray. Carbon dioxide (CO_2). Dry chemical. Chemical foam. **Unsuitable**

Extinguishing Media No information available

Flash Point > 250 °C / > 482 °F

Method - No information available

Autoignition Temperature Not applicable 607 °C / 1124.6 °F

Explosion Limits

Upper No data available
Lower No data available
Sensitivity to Mechanical Impact No information available
Sensitivity to Static Discharge No information available

Specific Hazards Arising from the Chemical

Dust can form an explosive mixture with air. Fine dust dispersed in air may ignite.

Hazardous Combustion Products

None known.

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear.

NFPA

Health	Flammability	Instability	Physical hazards
0	1	1	N/A

6. Accidental release measures

Personal Precautions Use personal protective equipment as required. Ensure adequate ventilation. Avoiddust formation.
Environmental Precautions Should not be released into the environment.

Methods for Containment and Clean Up Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

7. Handling and storage

Handling Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoidcontact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

Storage Protect from moisture. Store under an inert atmosphere. Keep container tightly closedinadry and well-ventilated place.

8. Exposure controls / personal protection

Exposure Guidelines This product does not contain any hazardous materials with occupational exposurelimitsestablished by the region specific regulatory bodies.

Engineering Measures Ensure adequate ventilation, especially in confined areas. Ensure that eyewashstationsand safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment.

Personal Protective Equipment

Eye/face Protection Wear appropriate protective eyeglasses or chemical safety goggles as describedbyOSHA's eye and face protection regulations in 29 CFR 1910.133 or EuropeanStandard EN166.

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure.

Respiratory Protection Follow the OSHA respirator regulations found in 29 CFR 1910.134 or EuropeanStandardEN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Hygiene Measures Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

Physical State Powder Solid
Appearance White
Odor vinegar-like
Odor Threshold No information available **pH** 7.5-9.2 5% aq.sol **Melting Point/Range** 324 °C / 615.2 °F **Boiling Point/Range** No information available **Flash Point** > 250 °C / > 482 °F **Evaporation Rate** Not applicable
Flammability (solid,gas) No information available **Flammability or explosive limits**
Upper No data available
Lower No data available
Vapor Pressure No information available **Vapor Density** Not applicable
Specific Gravity No information available **Solubility** 500 g/L (20°C)

Partition coefficient; n-octanol/water No data available **Autoignition Temperature** Not applicable 607 °C / 1124.6 °F
Decomposition Temperature No information available **Viscosity** Not applicable **Molecular Formula** C2 H3 Na O2
Molecular Weight 82.03

10. Stability and reactivity

Reactive Hazard None known, based on information available **Stability** Hygroscopic.

Conditions to Avoid Avoid dust formation. Incompatible products. Exposure to moist air or water. **Incompatible Materials**

Strong acids, Fluorine

Hazardous Decomposition Products None known

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information
Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Sodium acetate	LD50 = 3530 mg/kg (Rat)	LD50 > 10 g/kg (Rabbit)	LC50 > 30 g/m^3 (Rat) 1 h

No information available

Toxicologically Synergistic Products
Delayed and immediate effects as well as chronic effects from short and long-term exposure **Irritation** No information available

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as acarcinogen.

Component	CAS-No	IARC	NT	ACGIH	OSH	Mexic
Sodium acetate	127-09-3	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known
STOT - repeated exposure None known

Aspiration hazard No information available

No information available

Symptoms / effects,both acute and delayed

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

. Do not empty into drains.

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Sodium acetate	-	LC50: = 5000 mg/L, 24hstatic (Lepomis macrochirus)LC50: > 100 mg/L, 96hsemi-static (Danio rerio)	= 7200 mg/L EC50 Pseudomonas putida 18 h	EC50: > 1000 mg/L, 48h(Daphnia magna)

Persistence and Degradability Persistence is unlikely

Bioaccumulation/ Accumulation No information available.

Mobility . Will likely be mobile in the environment due to its water solubility.

Component	log Pow
Sodium acetate	-4.22

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classifiedasahazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT Not regulated
TDG Not regulated
IATA Not regulated
IMDG/IMO Not regulated

15. Regulatory information

United States of America Inventory

Component	CAS-No	TSCA	TSCA Inventory notification - Active/Inactive	TSCA - EPA RegulatoryFlags
Sodium acetate	127-09-3	X	ACTIVE	-

Legend:
TSCA - Toxic Substances Control Act, (40 CFR Part 710)
X - Listed
'-' - Not Listed

TSCA 12(b) - Notices of Export Not applicable

International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Australia (AICS), China (IECSC), Korea (ECL).

Component	CAS-No		DSL	NDSL	EINECS	PICC	EN	A	IECSCKECL
Sodium acetate	127-09-3		X	-204-823-8		X	X		X K

U.S. Federal Regulations

SARA 313 Not applicable

SARA 311/312 Hazard Categories See section 2 for more information

CWA (Clean Water Act) Not applicable

Clean Air Act Not applicable

Not applicable

OSHA - Occupational Safety and Health Administration

CERCLA Not applicable

California Proposition 65 This product does not contain any Proposition 65 chemicals. Not applicable

U.S. State Right-to-Know Regulations

U.S. Department of Transportation
Reportable Quantity (RQ): N
DOT Marine Pollutant N
DOT Severe Marine Pollutant N

This product does not contain any DHS chemicals.

U.S. Department of Homeland Security

Other International Regulations

Mexico - Grade No information available

16. Other information

Prepared By Health, Safety and Environmental Department

Email: tech@alfa.com

Creation Date 20-Jul-2009
Revision Date 14-Feb-2020
Print Date 14-Feb-2020
Revision Summary SDS authoring systems update, replaces ChemGes SDS No. 127-09-3.

Disclaimer
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End of SDS

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THECOMPANY/UNDERTAKING

1.1. Product identifier

Product Description: **Sodium thiosulfate pentahydrate, ACS** **Cat No. : 14518**
Synonyms Sodium Hyposulfite; Disodium Thiosulfate Pentahydrate.; DisodiumSalt Pentahydrate**CAS-No** 10102-17-7
Molecular Formula Na2 O3 S2 . 5 H2 O
Reach Registration Number -

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

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Company Alfa Aesar

Avocado Research Chemicals, Ltd.
Shore Road
Port of Heysham Industrial Park
Heysham, Lancashire LA3 2XY
United Kingdom
Office Tel: +44 (0) 1524 850506
Office Fax: +44 (0) 1524 850608

E-mail address uktech@alfa.com

www.alfa.com
Product Safety Department

1.4. Emergency telephone number

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Based on available data, the classification criteria are not met

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements

Hazard Statements

Precautionary Statements

2.3. Other hazards

No information available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS-No	EC-No.	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Sodium thiosulfate pentahydrate	10102-17-7		>95	-
Sodium thiosulfate	7772-98-7	231-867-5	-	-

Reach Registration Number	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Getmedical attention.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Get medical attentionimmediately if symptoms occur.

Ingestion Do NOT induce vomiting. Get medical attention.

Inhalation Remove to fresh air. If breathing is difficult, give oxygen. Get medical attentionimmediatelyif symptoms occur.

Self-Protection of the First Aider No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

No information available.

4.3. Indication of any immediate medical attention and special treatment needed **Notes to Physician** Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Water spray, carbondioxide (CO_2), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

Dust can form an explosive mixture with air. Containers may explode when heated. Thermal decomposition can lead toreleaseofirritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

Hazardous Combustion Products
Sulfur oxides, Sodium oxides.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) andfull protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures Ensure adequate

ventilation. Use personal protective equipment as required. Avoid dust formation.

6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. **6.3.**

Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoidingestion and inhalation. Avoid dust formation.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feedingstuffs. Donot eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, beforere-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

Class 13

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany)

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s):

Biological limit values
This product, as supplied, does not contain any hazardous materials with biological limits established by the region specificregulatory bodies

Monitoring methods
BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment ofexposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust **Derived No**

Effect Level (DNEL) No information available

Route of exposure	Acute effects (local)	Acute effects (systemic)	Chronic effects (local)	Chroniceffects (systemic)
Oral				
Dermal				
Inhalation				

No information available.

Predicted No Effect Concentration (PNEC)

8.2. Exposure controls

Engineering Measures
Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensurethateyewash stations and safety showers are close to the workstation location.

Personal protective equipment

Eye Protection Wear safety glasses with side shields (or goggles) (European standard - EN166) **Hand Protection** Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Natural rubber	See manufacturers recommendations		- EN 374 (minimumrequirement)	
Nitrile rubber				
Neoprene				
PVC				

Skin and body protection Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.
Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of thegloves. (Refer to manufacturer/supplier for information)
Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as thedangerof cuts, abrasion.
Remove gloves with care avoiding skin contamination.

Respiratory Protection When workers are facing concentrations above the exposure limit they must useappropriate certified

respirators.

Large scale/emergency use Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposurelimitsare exceeded or if irritation or other symptoms are experienced

Recommended Filter type: Particle filter 2

Small scale/Laboratory use Maintain adequate ventilation

Environmental exposure controls No information available.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State Solid

Appearance White
Odor Odorless
Odor Threshold No data available
Melting Point/Range 48.5 °C / 119.3 °F
Softening Point No data available
Boiling Point/Range No information available
Flammability (liquid) Not applicable Solid **Flammability (solid,gas)** No information available
Explosion Limits No data available

Flash Point No information available **Method -** No information available **Autoignition Temperature** No data available
Decomposition Temperature > 45°C
pH 6.0-8.4 10% aq. sol **Viscosity** Not applicable Solid **Water Solubility** 680 g/L (20°C)
Solubility in other solvents No information available
Partition Coefficient (n-octanol/water)
Component log Pow
Sodium thiosulfate -4.35
Vapor Pressure No data available
Density / **Specific Gravity** No data available
Bulk Density No data available
Vapor Density Not applicable Solid
Particle characteristics No data available

9.2. Other information

Molecular Formula Na2 O3 S2 . 5 H2 O
Molecular Weight 248.18
Evaporation Rate Not applicable - Solid

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Hygroscopic, Air sensitive, Light sensitive.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Avoid dust formation. Incompatible products. Excess heat. Exposure to moist air or water. Exposure to light. Exposure to air.

10.5. Incompatible materials

Strong oxidizing agents.

10.6. Hazardous decomposition products

Sulfur oxides. Sodium oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met **Dermal** No data available
Inhalation No data available

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;
Respiratory No data available
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable

Solid

Other Adverse Effects The toxicological properties have not been fully investigated. See actual entry inRTECSforcomplete information

No information available.

Symptoms / effects,both acute and delayed

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not containanyknown or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Do not empty into drains. Do not flush into surface water or sanitary sewer system.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Sodium thiosulfate pentahydrate	Pimephales promelas: LC50>10000mg/L/96h		

12.2. Persistence and degradability

Persistence Soluble in water, Persistence is unlikely, based on information available. **Degradability** Not relevant for inorganic substances.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Sodium thiosulfate	-4.35	No data available

12.4. Mobility in soil The product is water soluble, and may spread in water systems . Will likely bemobileintheenvironment due to its water solubility. Highly mobile in soils

No data available for assessment.

12.5. Results of PBT and vPvB assessment

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance **Ozone Depletion Potential** This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Chemical waste generators must determine whether a discarded chemical is classifiedasa hazardous waste. Consult local, regional, and national hazardous waste regulationstoensure complete and accurate classification.

Contaminated Packaging Empty remaining contents. Dispose of in accordance with local regulations. Donot re-useempty containers.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Waste codes should be assigned by the user based on the application for whichtheproductwas used.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

ADR Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

IATA Not regulated

14.1. UN number
14.2. UN proper shipping name
14.3. Transport hazard class(es)
14.4. Packing group

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
X = listed, Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), China (IECSC), Japan (ENCS), Australia(AICS), Korea (ECL).

Component	EINECS		ELI		TSC		DND	PIC	E	IE	AIC
Sodium thiosulfate pentahydrate	-	-		-	X						

Sodium thiosulfate	231-86			X							XKE-31 3

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export andimport of dangerous chemicals
Not applicable

National Regulations

WGK Classification See table for values

Component	Germany - Water Classification (VwVwS)	Germany - TA-Luft Class
Sodium thiosulfate pentahydrate	WGK1	
Sodium thiosulfate	WGK1	

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3

Legend

CAS - Chemical Abstracts Service **TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory **EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances **DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List **PICCS** - Philippines Inventory of Chemicals and Chemical Substances **ENCS** - Japanese Existing and New Chemical Substances **IECSC** - Chinese Inventory of Existing Chemical Substances **AICS** - Australian Inventory of Chemical Substances **KECL** - Korean Existing and Evaluated Chemical Substances **NZIoC** - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit **TWA** - Time Weighted Average **ACGIH** - American Conference of Governmental Industrial Hygienists **IARC** - International Agency for Research on Cancer **DNEL** - Derived No Effect Level Predicted No Effect Concentration (PNEC) **RPE** - Respiratory Protective Equipment **LD50** - Lethal Dose 50% **LC50** - Lethal Concentration 50% **EC50** - Effective Concentration 50% **NOEC** - No Observed Effect Concentration **POW** - Partition coefficient Octanol:Water **PBT** - Persistent, Bioaccumulative, Toxic **vPvB** - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road **ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association **IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code **MARPOL** - International Convention for the Prevention of Pollution from Ships **OECD** - Organisation for Economic Co-operation and Development **ATE** - Acute Toxicity Estimate **BCF** - Bioconcentration

factor VOC (volatile organic compound) **Key literature references and sources for data**
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE)andhygiene.

Prepared By Health, Safety and Environmental Department
Creation Date 06-Aug-2009
Revision Date 05-Jan-2021
Revision Summary SDS authoring systems update, replaces ChemGes SDS No. 10102-17-7/1.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No1907/2006

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information andbelief at thedate of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,transportation, disposal and release and is not to be considered a warranty or quality specification. The informationrelates only to the specific material designated and may not be valid for such material used in combinationwithanyothermaterials or in any process, unless specified in the text

End of Safety Data Sheet



SAFETY DATA SHEET

Creation Date 26-Sep-2009

Revision Date 22-Dec-2020 **RevisionNumber**5

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THECOMPANY/UNDERTAKING

1.1. Product identifier

Product Description: Starch, soluble
Cat No. : 424490000; 424490020; 424490025; 424490100; 424491000; 424495000 Synonyms amylodextrin; starch; a-Amylodextrin
CAS-No 9005-84-9
EC-No. 232-686-4
Molecular Formula (C6 H10 O5)n

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company UK entity/business name

Fisher Scientific UK
Bishop Meadow Road, Loughborough,
Leicestershire LE11 5RG, United Kingdom

EU entity/business name
Acros Organics BVBA
Janssen Pharmaceuticalaan 3a
2440 Geel, Belgium

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Based on available data, the classification criteria are not met

Environmental hazards
Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements

Hazard Statements

Precautionary Statements

2.3. Other hazards

No information available

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS-No	EC-No.	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Soluble Starch	9005-84-9	EEC No. 232-686-4	1	-

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Getmedical attention.

Skin Contact Wash off immediately with soap and plenty of water while removing all contaminatedclothes and shoes. Get medical attention.

Ingestion Clean mouth with water. Do NOT induce vomiting. Get medical attention.

Inhalation Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. If symptoms arise, call a physician.

Self-Protection of the First Aider No special precautions required.

4.2. Most important symptoms and effects, both acute and delayed

No information available.